

Two-Way Log-Budget Decomposition of Prime-Shell Operator Interaction

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Abstract

TPC-305 replaced one shell’s source-first target by an aligned neighboring label while holding that shell’s physical operator fixed. The resulting orientation atlas isolated a within-row target comparison but left the two-row operator interaction implicit. We arrange the four positive budget cells as an operator-by-target table and define two log target-switch effects. Their mean is a target-main contrast m , their half-difference is an operator interaction contrast i , and the exact identity $m^2 - i^2 = d_L d_R$ gives a strict dominance test. Replaying all 18 TPC-305 cases at three source normalizers yields 54 derived rows: target-main dominance occurs in 12 of 18 cases and interaction dominance in 6 of 18, with no unresolved case. At the central $Q = 60 \rightarrow 70$ fracture the split is 5/6 versus 1/6, and all three inherited same-prefix cases are target-main dominant. The finite diagnostic quantifies, but does not remove, the common-ambient and causal-identification obstruction.

1 From a target swap to a two-way table

TPC-305 is the immediate predecessor of this report [1]. It holds each physical shell/operator fixed, changes the target only on an overlap using the aligned neighboring sign label, and retains the native sign off the overlap. On the fixed $(N, H, z) = (512, 58, 5)$ spine with $Q = 50, 60, 70, 90$, it finds five of six central cases in which the right-neighbor target is cheaper on both fixed operators. That statement is useful, but two operators still have different physical output rows. A single orientation label therefore hides an interaction between operator identity and target identity.

We retain exactly the TPC-305 finite data: two kernel exponents, three relative RMS tolerances, three adjacent shell pairs, and three positive source normalizers. No new shell or asymptotic regime is introduced. For one pair, write the four normalized budget cells as

	left target	right target
left operator	B_{LL}	B_{LR}
right operator	B_{RL}	B_{RR}

(1)

Here “left target” means the native left label on the left row and its aligned transported completion on the right row; “right target” has the opposite completion. Thus (1) is a protocol-level holdout table, with the same shell-specific completion convention as TPC-305.

2 Effects and exact decomposition

The common intervention is the target switch left $\rightarrow$ right. Its row-wise log effects are

$$d_L = \log \frac{B_{LR}}{B_{LL}}, \quad d_R = \log \frac{B_{RR}}{B_{RL}}. \quad (2)$$

Table 1: Two-way dominance census over six cases per transition.

transition	target-main	interaction	same-prefix main
50 → 60	4	2	0
60 → 70	5	1	3
70 → 90	3	3	0

TPC-305 recorded $R_L = B_{LR}/B_{LL}$ and $R_R = B_{RL}/B_{RR}$, so $d_L = \log R_L$ and $d_R = -\log R_R$. Define

$$m = \frac{d_L + d_R}{2}, \quad i = \frac{d_L - d_R}{2}, \quad q = \frac{|i|}{|m|}. \quad (3)$$

The quantity m measures the target-switch effect averaged over the two operator rows; i measures how much that effect changes between rows.

Proposition 1 (Two-way squared-contrast identity). *For positive cells in (1),*

$$d_L = m + i, \quad d_R = m - i, \quad m^2 - i^2 = d_L d_R. \quad (4)$$

If both row effects are nonzero, equal signs are equivalent to $|m| > |i|$, and opposite signs are equivalent to $|i| > |m|$.

Proof. Adding and subtracting (3) gives the first two identities. Expanding the difference of squares gives

$$m^2 - i^2 = \frac{(d_L + d_R)^2 - (d_L - d_R)^2}{4} = d_L d_R.$$

The product on the right is positive exactly for equal nonzero signs and negative exactly for opposite signs. Taking square roots gives the two strict inequalities. $\square$

Proposition 2 (Row-normalization invariance). *Independent positive rescalings of the two rows of (1) leave d_L, d_R, m, i and q unchanged.*

Proof. Multiplying both cells in one row by the same positive factor cancels in its ratio in (2). Apply this separately to the two rows. $\square$

We call $|m| > |i|$ target-main dominance and $|i| > |m|$ interaction dominance. The sign of the two effects distinguishes the right-neighbor and left-neighbor main preferences. This terminology is algebraic; it is not a causal claim.

3 Finite certificate

The producer reads the locked TPC-305 ratio intervals and applies 80-digit logarithmic interval arithmetic with a padding of 10^{-30} . The independent checker does not import this producer: it replays the logarithms with 100-digit Decimal arithmetic, verifies all 54 enclosures, checks (4) on the printed centers, and reconstructs the case census. A stress suite exhausts 625 positive four-cell tables and independent positive row scalings.

Table 1 gives the case-level result. A main-dominant case has same-sign row effects; an interaction-dominant case has opposite signs.

The central six rows are displayed in Table 2. The numbers are interval centers; the certificate stores 38-digit endpoints. The three normalizers agree on every strict class.

Across all 36 target-main normalizer rows, the upper endpoint of q is below 0.88. Across all 18 interaction-dominant normalizer rows, the lower endpoint is above 1.2. For the three central same-prefix cases the maximum upper endpoint is below 0.64. Thus the finite atlas has a visible gap between the two algebraic regimes rather than a borderline classification.

Table 2: Central log effects and contrasts (interval centers).

e	τ	m	i	q	dominant term
1	.25	-.203481	.500980	2.46205	interaction
1	.50	-.209604	-.133643	.637596	target-main
1	.75	-1.27333	-.629188	.494130	target-main
2	.25	-1.22433	-1.07622	.879029	target-main
2	.50	-2.20188	-.636939	.289271	target-main
2	.75	-3.19790	-1.85187	.579089	target-main

4 What this does and does not identify

The positive result is a compact interaction diagnostic. TPC-305’s central five-of-six right-label signal is exactly the five-of-six target-main regime in the table, and the same-prefix parent descents remain target-main dominant. The identity (4) explains why: both row-wise target effects have the same sign, so their mean is larger in magnitude than their half-difference.

The obstruction is not erased. Six cases have opposite-sign row effects, so operator interaction is larger than the mean target contrast; one of these is at the central fracture and three occur on the final transition. Moreover, the right and left targets in (1) use native off-overlap completions specific to their row. Consequently the table is not a balanced intervention on a single common ambient target vector. A common union-shell completion and an operator holdout are required before interpreting m as a causal effect.

The decomposition also supplies no arithmetic L^2 estimate, fixed power saving, full Route-B Gate-B theorem, or twin-prime result. The Session-named Route-A/Route-B evaluator files are absent from this checkout, so the claim level is the local fail-closed level recorded in the route-evaluation note.

Reproducibility and next question

The canonical certificate is `results/tpc306_certificate.json`. The producer, independent checker, stress suite, and Bridge-B checker are included with the project. The next minimal question is whether the dominance gap survives a common-ambient union-shell completion and alternative off-overlap transports. No claim is made that the finite orientation persists as Q or N grows.

References

- [1] Liang Wang. A fixed-operator counterfactual for overlapping prime-shell labels, 2026. Local TPC-305 release in the prime-dynamics research repository.